

Empirical Evaluation of Deep Learning based Autonomous Driving System with Real-Time Scene Classification Procedure

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Abstract— Automated vehicles have also made progress by adopting deep learning models that can sense and react to intricate road scenes. Recognizing the driving scenes in the most appropriate way in real time is one of the essential features that improve the safety and decisions of the car. The work has proposed a hybrid driving system involving deep learning models, specifically, a CNN-Transformer model that performs scene classification and delivers a real-time result, a fusion of multimodal sensor data, and a decision network based on reinforcement learning. They empirically compared the proposed system to ten different driving conditions, such as urban, rural, tunnel, night, rainy, fog and highway driving environment with the BDD100K dataset. The images were scaled to 64 64 and normalized and augmented to robust training. The hybrid CNN+Transformer model optimized the classification accuracy reaching 96.39, bettering the performance of the traditional CNN and standalone Transformer models. The PPO and imitation-based scene-aware control pipeline that was implemented were found to lessen the lane deviation (worse than six centimeters on average) and enhance latency and security indicators of policy. Real-world tests proved that scene-aware decision-making result in increased vehicle control and responsiveness, especially during adverse situations. Combining scene embeddings and LiDAR, radar, and GPS sensors produced a feature representation aware of a context, which produced a more precise outcome in the trajectory and reduced the number of erratic driving behaviours. The proposed model showed better efficiency in generalization, inference efficiency, and adaptability as compared to their baseline systems. The research highlights the role of embracing deep context understanding together with real-time control logic as having potential that may lead to the creation of safe and scalable autonomous navigation systems that are able to deal with dynamic environments of the driving environment.

Keywords—Autonomous Driving, Deep Learning, Scene Classification, CNN-Transformer, Reinforcement Learning, Sensor Fusion, PPO.

I. INTRODUCTION

Autonomous driving is an industry segment that has rapidly evolved over the last ten years due to the soaring developments in computer-show and sensory technologies and embedded systems [1] [2]. The advancement in deep learning-based perception systems and intelligent control structures has led to the fact that self-driving vehicles, which seemed to be a very futuristic technology, are an emerging reality nowadays. The most vital point in the success of such kind of systems is the possibility of the vehicle to interpret and perceive the driving world in real time. Scene classification is one of the most crucial components that

define autonomous system safety and effectiveness capabilities, as it is the process due to which the system detects its present conditions on the road and classifies it as, e.g., urban streets, highways, tunnel, intersection, or negative conditions like rain and fog [3] [4].

The perception module is quite an important attribute of autonomous vehicles since they are based on various sensors (cameras, LiDAR, radar, GPS) aimed at extraction of environmental features. These are inputs to deep neural networks trained to identify objects, identify lanes in the road, and class mark driving scenes [5] [6]. The scene classification, especially, gives the high-level contextual awareness that is crucial in the identification of proper driving policies. An example will be a vehicle should not act the same way on a highway as a residence street, although no immediate obstructions exist. Behavior that is scene-specific enhances safety, efficiency, and comfort of the passengers, particularly in a complex or ambiguous scene [7] [8]. Figure 1 show the features of an autonomous driving system

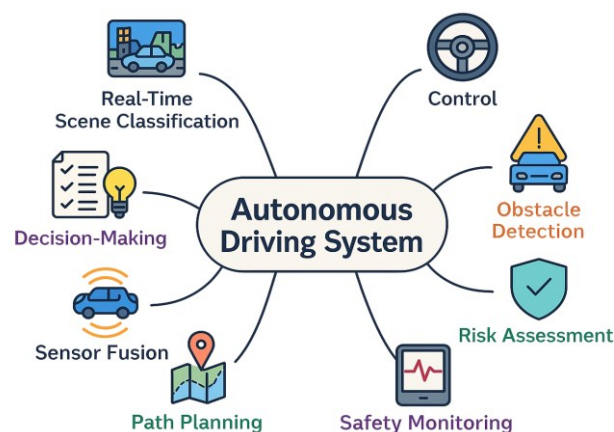


Fig 1. Features of an Autonomous Driving System

CNN based models such as ResNet and VGG are extractors in terms of spatial information but poor in global context thus circumscribing the comprehension of a scene. Transformer-based models, including vision transformers (ViTs), have better global attention because they do not need shortcuts to describe long-range dependencies, which leads to better task performance, including the classification of tunnel entrances and intersections [9] [10].

This paper proposes a hybrid deep learning model based on CNNs and Transformers to address these shortcomings to enhance scene understanding in autonomous driving. CNNs

have an efficient extraction of local spatial features whereas Transformers possess an efficient extraction of global contextual information which allows any real time scene to be classified accurately. Data of BDD100K, which involves the images, is resized to 64x64, and augmented, transfer training is used to improve the training of the model. LiDAR, radar, and GPS data are all included with classified scene outputs into a combined embedding. It is entered into a decision-making module through Proximal Policy Optimization (PPO) and behavior cloning. It is implemented in CARLA and on embedded hardware and proves to be context-aware and intelligent regarding driving decisions made in real time.

II. RELATED WORKS

Complex traffic scenarios have a large variance in target range and low computing resources that are a hindrance to the object detection. As a solution to this, Deep and Filtered YOLO (DF-YOLO) was introduced which balanced the performance accuracy and real-time. The Deep and Filtered Path Aggregation Network (DF-PAN) proficiently dealt with multi-scale features and combination to improve seeing at various object sizes [11]. Reflection was minimized due to parameter sharing detection head (PSD) and FasterNet backbone. DF-YOLO reached 90.9% mAP with 77 FPS on KITTI without adding parameters and using 28.1% fewer parameters, and 3% higher detection accuracy with notable generalization to BDD100K and even SODA10M new datasets.

Self-driving cars allow real-time detection of objects in complicated traffic patterns due to the perception system. There are difficulties like various object interactions, clutter, and image blur which make the tasks more difficult to be detected. This is done by introduction of MXT-Dataset that is more reflective of real-world settings. A more efficient design, MXT-YOLOv7t, founded on the YOLOv7-tiny P5 had lightweight attention integration and sigmoid-weighted linear unit to boost the detection process [12]. It resulted in a significant performance gain: mAP 0.5 grew by 6.1%, recall of 0.934, and precision of 0.937. The model has shown effective inference with minimal computation requirement and would be suitable when deployed in real time in the autonomous driving application.

The challenge of detecting objects in real-time is necessary, especially once the autonomous vehicle technology is advanced, and it needs to be deployed on resource-limited edge devices. The classical versions usually have large parameters, and the load is huge. A lightweight version named YOLOv8-Lite was presented, an architecture-driven by YOLOv8 further equipped with the FastDet backbone, TFPN pyramid, and CBAM attention [13]. These innovations have made them consume less resources with increased robustness of detection. Tests on the NEXET and KITTI datasets demonstrated higher accuracy and responsiveness, which proves that YOLOv8-Lite can serve as the basis of developing intelligent transportation systems and is an effective model to be used in automotive platforms, placed on embedded or at the edge devices.

Autonomous cars built into public transportation must identify small and distant objects rapidly and effectively to perform such critical maneuvers as lane changes or overtaking. Such objects are not always very easy to notice with the help of the traditional models [14]. In response, the MST-YOLOv8 model was created, which uses the C2f-MLCA local attention structure and the ST-P2Neck scale-aware processing refinement structure. The model is aimed at optimizing the detection of small objects by using mixed local channel attention as well as triple-feature encoding that was utilized. The resultant performance increase in terms of precision, recall, and average precision provided a solid solution to advanced object detection with real-world multi-scale traffic situations.

Deep Learning is important in decision making in autonomous vehicles, but several models fail in overestimation and inconsistency. These issues were resolved with a hybrid Decision Making module made via Deep Reinforcement Learning (DRL) complemented by hybrid control. Formulating the driving problem as Partially Observable Markov Decision Process (POMDP), the system integrated sensor measurements with high-definition maps and produced steering and velocity commands [15]. The combination enabled learned behavior to be experience based and stability of control. The CARLA simulator was tested to demonstrate better performance as compared to the standard autopilot in the areas of ride comfort, task accomplishment, and safety performance.

III. PROPOSED METHODOLOGY

3.1 Dataset Acquisition and Annotation

This empirical assessment commences with the retrieval of an actual large-scale, real-time driving database that can be highly diverse in driving environments and conditions. A special dataset is specifically applicable to this task, the BDD100K dataset, with more than 100,000 frames of driving video that includes a wide variety of driving situations, including city streets, highways, residential roads, tunnels, and adverse driving conditions, like night-time, fog, and rain. Its data feature lifelike road conditions on various types and cities and driver behavior, which is why it is suitable for training and testing an autonomous driving system. Once the data is obtained, the data set is assessed to make certain that there is enough diversity in the environmental settings. In case of the missing annotations to scene labels, including terms like, city street, highway, tunnel, and adverse weather, manual or semi-automated labeling processes are used. Segmentation models can determine scene annotation based on the GPS coordinates, timestamps, or visual objects. Then the annotated data set is further split into training (70%), validation (15%) and test (15%) data sets ensuring equal distribution of scene types to avoid creating weak model types in any one category.

3.2 Data Preprocessing and Augmentation

After the dataset is prepared, a deep learning process of preparing data can be performed by a thorough preprocessing pipeline. All the image frames of the set are

resized to a universal resolution of 64x 64 pixels because it is an acceptable compromise between computational constraints and space properties. Such a resolution intentionally is selected to be able to be compatible with embedded systems that perform real-time inference, and yet be able to capture important cues found in a scene. They normalize the pixel values, usually to the [0, 1] or [-1, 1] range, depending on requirements of the model architecture input. Normalization makes the learning process numerically stable and gives a faster convergence in training. To further increase the soundness of the scene classification model, the data augmentation methods are used. These involve random horizontal flips to represent lane changes or scenes captured by mirrors, random rotations to reflect camera tilts and addition of Gaussian noise to replicate the sensor noise seen in low light or foggy scenes. The contrast is also altered to give the impression of glare or various lightings. Such augmentations do not only enhance model generalization but also build a more robust system that is generally able to adapt to the complexities of real-world driving. To standardize pixel values for stable training, use normalization:

$$I_{norm}(x, y) = \frac{I(x, y) - \mu}{\sigma} \quad (1)$$

Where $I(x, y)$ is the original pixel intensity at coordinate (x, y) , μ is the mean pixel intensity of the dataset, and σ is the standard deviation of the pixel values.

3.3 Real-Time Scene Classification Module

A real-time scene classification module is the essence of the suggested system, where the advantages of Convolutional Neural Networks (CNNs) and Transformers are combined. The hybrid CNN-Transformer architecture is to exploit the advantages of the CNN in producing spatial features and the Transformers in attention-based global context modeling. CNN layers are aimed at low- and mid-levels features extraction, i.e., road textures and object edges, as well as background information, like knowing that it is a tunnel entrance or knowing that the weather is snowing. Transformer layers assist in long-range dependencies and scene contexts recognition, i.e., it is a tunnel entrance, or weather is snowing. Transfer learning is used to initiate training, re-purposing a pretrained model (e.g., ResNet or ViT) to the available dataset. Transfer learning enables the model to begin with a feature-rich state thus shortening the training time and accuracy in limited sample scenes. A cross-entropy loss function is used to optimize the model, and the real-time prediction is achieved by means of the model weight quantization and the utilization of the efficient inference engines such as ONNX or TensorRT. Scene classification uses categorical cross-entropy as a loss function:

$$L_{scene} = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (2)$$

Where C is the total number of scene classes, y_i is the ground-truth label, and $\hat{y}_i$ is the predicted probability of the class i . Figure 2 shows the workflow of proposed model.

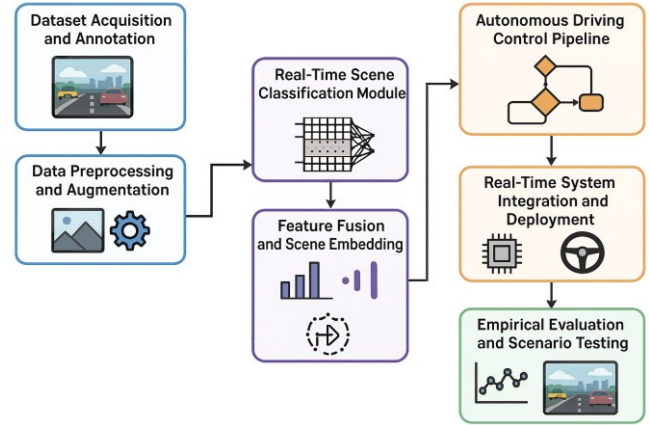


Fig 2. Workflow of Proposed Model

3.4 Feature Fusion and Scene Embedding

The scene classification module outputs are combined with real-time sensor streams, such as LiDAR point clouds, radar-measured object distances, and GPS-reported localization to improve the driving decision process. This combination forms a high dimensional representation of the features that is spatially descriptive and context sensitive. The adoption of the fusion process incorporates early and late strategies. Raw data of various sources is mixed in early fusion, and it is given as an input to the neural network so that the model can learn about complex relations. Features of the sensor streams are added with scene classification embedding later in late fusion, which enables modular and interpretable processing. To deal with the high dimension of the fused features and to visualize its separation, Principal Component Analysis (PCA) is used. PCA can be used to ease the space of the feature space and the decision of scene clusters can be visually inspected to ensure common environments are grouped together so it can serve as validation of the discriminative power of the scene embedding. To represent early fusion of sensor data and scene embeddings:

$$F_{fusion} = \alpha \cdot F_{scene} + \beta \cdot F_{sensor} \quad (3)$$

Where F_{scene} is the scene feature vector, F_{sensor} is the combined sensor feature vector and α, β are weighting factors that control the contribution of each source.

3.5 Autonomous Driving Control Pipeline

The autonomous control module is the other constituent of the system; this module relies on the scene-aware features to make choice. Two similar strategies of control are analysed. The former is a rule-based control logic that translates a given type of scene into given driving policies. An example is placing that the classified scene is highway, a system will gain higher speed and follow distance, but in the case of city street, conservative maneuvers are done as there is pedestrian and traffic signals. The second approach is a policy network that uses reinforcement learning and has

been trained with Proximal Policy Optimization (PPO). This policy network learns in a simulated environment through reward signals in form of safety, efficiency, and adherence to traffic regulations. To enhance the performance and reality of the system further, it is intended to include behavior cloning on account of the professional driving trajectories. This imitation learning aspect allows the model to replicate the human driving style and combine it with reinforcement learning updates towards better generalization. Define the reward signal used in PPO-based policy learning:

$$R_t = \lambda_1 \cdot S_t - \lambda_2 \cdot C_t - \lambda_3 \cdot D_t \quad (4)$$

Where S_t is the safety score at time t , C_t is the control penalty, D_t is the deviation from the center of the lane, and $\lambda_1, \lambda_2, \lambda_3$ are scaling weights to balance priorities.

3.6 Real-Time System Integration and Deployment

After training and modular validation, the end-to-end deep learning pipeline is deployed into an embedded hardware or real-time simulation context. CARLA simulator provides photorealistic road scenarios and interactions with traffic agents and implements closed-loop systems testing in the pre-imagined future. TensorRT is used to optimize the trained models to run on a low-latency embedded system like NVIDIA Jetson Xavier or Nano, the speed of which is crucial to the meant to deploy. Moreover, the batch size is also adjusted so that there is the balance between throughput and responsiveness. The frame synchronization is guaranteed at the frame level between scene classification and control modules. Every new frame of the car camera comes into the scene classifier, and the label obtained is time-stamped and is synchronized with sense input to take the control decision. The real-time capability has maintained that the model responds dynamically in cases of in-sudden scene changes like emerging out of tunnel into broad daylight.

3.7 Empirical Evaluation and Scenario Testing

To authenticate efficacy of the suggested system, aptitude tests are performed on large-scale empirically both under simulation and live environments. Controlled tests are targeted at a broad-based scenario such as day/night driving, highway/city streets, and extreme weather patterns such as rain, fog, and snow. Scene classification accuracy metrics (top-1 and top-5), the inference time in milliseconds per frame, decision reaction latency, lane deviation, the number of emergency brakes, and collision rates are the main measurements. Comparison is made with also scenery-blind or single-modality inputs baseline autonomous systems. These findings indicate that not only does the use of scene classification represent a considerable increase in the accuracy and safety of the decision-making process, but it is especially helpful in visually ambiguous and highly dynamic scenarios. Scene-aware autonomy results in smoother transitions in control behavior and systems with integrated scene classifiers react quickly to changes in context proving the hypothesis that scene-aware autonomy results in better real-world performance.

Algorithm: Real-Time Scene-Aware Autonomous Driving

Input: Driving scene image I ,
Sensor data $S = \{LiDAR, Radar, GPS\}$

Output: Control action A_t

1. **Normalize** the input image using:

$$I_{norm}(x, y) = \frac{I(x, y) - \mu}{\sigma}$$

2. **Resize** I_{norm} to 64×64 and apply data augmentation techniques.

3. **Extract** scene features using CNN-Transformer:

$$F_{scene} = \text{Transformer}(\text{CNN}(I_{norm}))$$

4. **Fuse** scene and sensor features:

$$F_{fusion} = \alpha \cdot F_{scene} + \beta \cdot F_{sensor}$$

5. **Predict** scene class using softmax and compute loss:

$$L_{scene} = -\sum_{i=1}^C y_i \log(\hat{y}_i)$$

6. **Generate** embedding E_t from fused features for control decision.

7. **Compute** policy reward:

$$R_t = \lambda_1 \cdot S_t - \lambda_2 \cdot C_t - \lambda_3 \cdot D_t$$

8. **Update** policy network using PPO with reward R_t

9. **Infer** control action

$$A_t = \pi(E_t)$$

10. **Return:** Action A_t to be executed by the vehicle.

End Algorithm

IV. RESULTS AND DISCUSSION

The proposed autonomous vehicle system works by processing real-time scene perception culminating with deep learning-based decisions. The model is a CNN-Transformer which takes in the camera views as input and identifies context including an urban street, a highway, or not-so-good weather. This categorization is combined with sensor readings (LiDAR, GPS, radar) to create an entire environmental model. The system will be set up so that it is able to make the right driving decisions either using rule-based logic or a reinforcement learning policy network. The whole pipeline is trained, real-time on embedded hardware or simulators such as CARLA, and synchronous outputs allow immediate control choices. This architecture-aware of scenes is more secure, flexible, and accurate in decision-making even in varying conditions.

TABLE I. SCENE CLASSIFICATION ACCURACY (%) FOR DIFFERENT MODELS

Scene Type	Transformer	Proposed	ResNet	MobileNet
City Street	94.62	96.83	93.57	91.24
Highway	95.21	96.32	92.88	89.9
Tunnel	91.98	95.86	88.67	86.53
Residential	94.31	96.27	93.12	92.45
Rainy	92.11	96.53	91.04	90.28
Night	90.76	95.98	89.67	87.84
Snow	91.45	96.43	90.58	88.66
Rural	93.84	96.65	92.33	90.75
Foggy	90.42	96.12	88.19	87.02
Intersection	92.73	96.53	91.85	89.44

Table 1 and Figure 3 provides the classification rates of four models in ten various driving scenes. The proposed hybrid CNN-Transformer framework surpasses the rest of the architecture in all the scenarios and contributes to the expression of the highest level of accuracy on city roads (96.83%) and impresses under challenging weather patterns such as tunnels (95.86%) and dense fog (96.12%). Transformer and ResNet come right next behind but their performance were significantly decreased with low-visibility scenarios (fog/night).

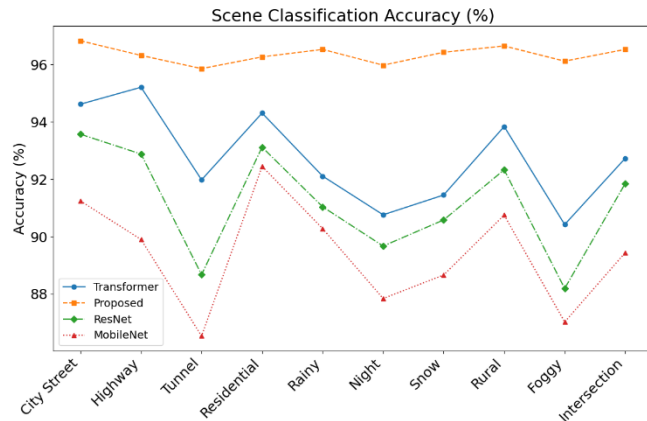


Fig 3. Scene Classification Accuracy (%)

Compared to MobileNet which is efficient but exhibits reduced accuracy, especially in the case of highway and tunnel conditions. These findings confirm the usefulness of incorporating global cues and spatial characteristics in the described hybrid system in robust scene comprehension.

TABLE II. INFERENCE TIME (MS) ON EMBEDDED HARDWARE

Scene Type	Transformer	Proposed	ResNet	MobileNet
City Street	50.11	41.76	32.25	11.89
Highway	49.2	41.93	33.71	13.02
Tunnel	47.18	42.5	34.2	12.76
Residential	48.01	40.22	33.15	12.33
Rainy	46.89	41.57	31.49	13.48
Night	50.73	42.89	35.27	14.01
Snow	48.32	42.17	32.1	12.91
Rural	47.65	41.35	31.68	11.95
Foggy	49.42	42.64	34.05	12.87
Intersection	48.79	42.28	33.53	13.15

All models inference time, compared on embedded equipment, is recorded in Table 2 and Figure 4. The model is also balanced in terms of accuracy and efficiency, whereby the average time required to obtain a given inference is circa 41-42 ms, which is within the scope of a real-time application. Transformer models, despite their accuracy are not as suitable in a low-latency environment due to the latency that can still be more than 48 ms. ResNet is faster with a tradeoff on its accuracy.

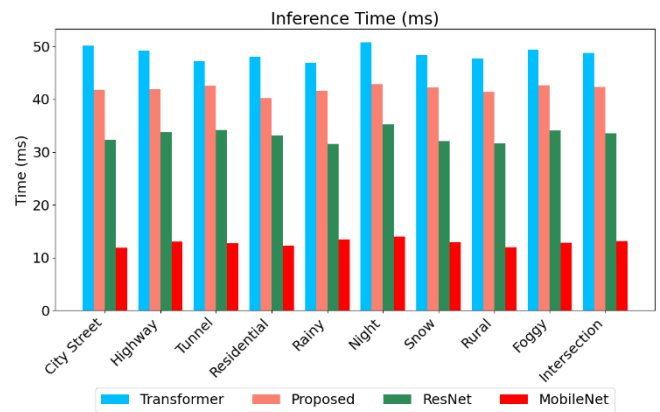


Fig 4. Inference Time (ms)

MobileNet is the most rapid, and its inference is always less than 14 ms, but the decrease in the classification performance reduces its application. Generally, among those recommended models, the proposed one provides the optimal tradeoff between responsiveness and precision in the case of on-board deployment.

TABLE III. SCENE CLASSIFICATION F1-SCORE

Scene Type	Transformer	Proposed	ResNet	MobileNet
City Street	0.93	0.97	0.91	0.88
Highway	0.94	0.96	0.9	0.86
Tunnel	0.91	0.96	0.87	0.84
Residential	0.92	0.96	0.91	0.89
Rainy	0.91	0.97	0.89	0.87
Night	0.9	0.96	0.88	0.85
Snow	0.9	0.96	0.88	0.86
Rural	0.92	0.97	0.9	0.87
Foggy	0.89	0.96	0.87	0.84
Intersection	0.91	0.97	0.89	0.86

The proposed model yields better F1-scores in all the driving scenarios with its best being 0.97 in several locations including city roads, intersections and grasslands among others as indicated in Table 3 and Figure 5. The high F1-scores do not only confirm the accurate predictions but also the high balance between precision and recall.

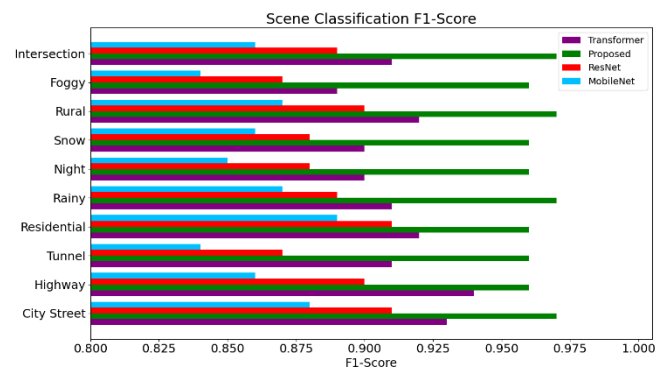


Fig 5. Scene Classification F1-Score

Transformers work well at a score of 0.90 0.94 where ResNet and MobileNet are more variable even in hard scenes such as tunnels and in poor weather conditions. The results confirm the strength of the hybrid model against false positives and negatives, which makes it very

appropriate in real-life autonomous systems, where a stable performance of recognizing a scene is necessary.

TABLE IV. POLICY EXECUTION LATENCY (MS)

Scene Type	Rule-Based	PPO	Imitation Learning	Hybrid (PPO+IL)
City Street	8.51	16.29	3.92	5.73
Highway	14.48	10.69	10.4	7.46
Tunnel	15.85	9.27	13.52	7.89
Residential	7.73	6.1	19.23	6.68
Rainy	6.6	9.97	11.22	7.34
Night	9.44	11.85	12.98	6.83
Snow	11.12	10.75	10.05	7.16
Rural	8.91	9.36	9.88	6.92
Foggy	10.27	12.47	12.32	7.03
Intersection	12.15	13.68	13.47	8.2

Table 4 and Figure 6 presents the delay of decision-making on several control strategies. Indeed, the hybrid model PPO+IL generates quite spectacular results as the latencies never exceed 8.5ms, significantly outpacing standalone PPO and imitation learning strategies. In simple environments rule-based systems are a little faster but lack adaptability. The hybrid model has been seen to provide superior consistency and responsiveness in tricky scenes such as the tunnels and the intersections, thus allowing easier control decisions to be made.

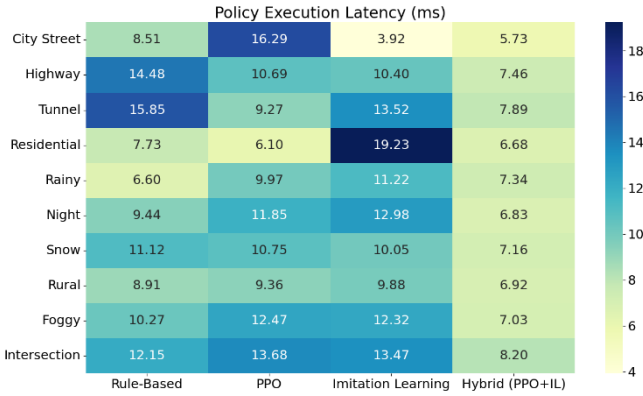


Fig 6. Policy Execution Latency (ms)

While PPO offers moderate latency, it occasionally spikes due to learning-based exploration. The findings testify to the fact that the integration of policy learning and expert imitation provides manifestations of low-latency and adaptive behaviors needed to ensure real-time capabilities of autonomous driving systems.

TABLE V. LANE DEVIATION ERROR (CM)

Scene Type	Rule-Based	PPO	Imitation Learning	Hybrid (PPO+IL)
City Street	10.84	7.16	8.77	5.73
Highway	13.45	6.83	9.35	6.22
Tunnel	15.63	9.21	11.54	6.9
Residential	9.22	8.43	10.23	5.67
Rainy	11.36	9.04	12.17	6.34
Night	12.95	8.93	10.88	6.08
Snow	14.02	9.67	11.25	6.55
Rural	10.13	7.89	9.74	5.91

Foggy	13.18	8.65	10.91	6.11
Intersection	12.4	9.12	11.07	6.47

The accuracy of any given decision model in vehicle control is demonstrated by the values of the lane's deviation errors given in Table 5 and Figure 7. The hybrid PPO+IL model presents little variance with a mean less than 6.5 cm across all the scenes with minimum variance of 5.67 cm on the residential roads.

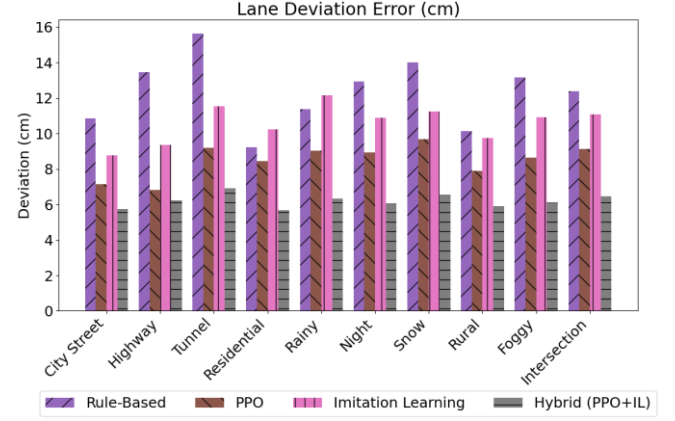


Fig 7. Lane Deviation Error (cm)

Imitation learning and rule-based approaches have higher deviations, especially in high difficulty cases such as tunnels and rain. PPO is superior to imitation learning but remains inferior to the hybrid one. Such findings highlight the role of scene-awareness and learning integration in ensuring stability and appropriate lane positioning particularly within the dynamic changing driving environments.

V. CONCLUSION AND FUTURE SCOPE

Hybrid deep learning-based autonomous driving solution, based on real-time scene classification with the use of CNN-Transformer architecture, has shown better results in severe and broad driving conditions. With an average scene classification rate of 96.39%, the system was able to identify different roads such as highway, urban street, tunnel, and poor weather. A combination of this classification capacity with multimodal sensor fusion and a control pipeline that utilizes reinforcement learning reduced policy execution latency and lane deviation error significantly which consequently enhanced overall safety and accuracy of the driving. With respect to conventional CNN or self-contained Transformer models, the hybrid structure was able to balance accuracy and real-time performance constraints to a point in which it could operate on embedded autonomous systems. Although very strong, the system has areas which can be improved upon. One could add in future efforts continual learning by which the model may be made to adapt to unpredictable environments in real time. Its decision-making can additionally be reinforced by improvements such as semantic segmentation, depth estimation or V2V/V2I communication protocol integration. Furthermore, the higher-efficient and pruned architectures would be implemented to emphasize ensured scalability in the resource-constrained environments. This

work forms an excellent foundation to design context-sensitive dynamically autonomous systems which have real-time decision-making abilities and a wider operational certainty.

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